

filtering-application(): the same data filtering machinery can be used for different filtering tasks

=> def language-identification(): find text of a specific language

①: off-the-shelf classifier
②: supports 176 languages

(e.g. Chinese
(German))

③: Trained on multilingual site: wikipedia, Tatoeba and
SE Times

example: Polma keeps pages with $p(\text{English}) \geq 0.5$

step 1: Download the model from url

step 2: Make predictions: English / French / German / C++
Spanish.

cautions:

Difficult for short sequences

Difficult for low-resource languages

could accidentally filter out dialects of English

Hard for still similar languages (Malay and Indonesian)

Not defined for code-switching (e.g. Spanish + English)

=> def quality-filtering():

① no use of model-based filtering: C4, Retineweb, Findex

②: use of model-based filtering: GPT3, Llama, DCLM
Became the norm

examples: GPT3: { positives: samples from wikipedia, webText, (Books)
negatives: samples from common crawl

Llama: { positives: from pages referenced by wiki
negatives: from common crawl.
similar

keep docs that are classified positive

⇒ phi-1: really high-quality data (textbooks) to train a small model (1.5B)
includes synchro data from GPT-3.5 (later GPT-4) and filtered data

⇒ def toxicity-filtering():

example: Toxicity filtering in Polma. ~~*~~

Dataset: Jigsaw Toxic comments data (2018)

- ①: project goal: help people have better discussions online
- ②: data: comments on wikipedia talk page annotated with {toxic, severe-toxic, obscene, threat, insult, identity-hate}

Trained 2 fastText classifiers:

①: hate: positive = {unlabeled, obscene}, negative = all else

②: NSFW: positive = {obscene}, negative = all else

↓
not safe words

step1:

download model from url: model = fasttext.load_model('model.pth')

step2: make predictions: predictions = model.predict('I "stentent"')

Def duplicates():

- 2 types:
 - ①: Exact duplicates (mirror sites, Github forks)
 - ②: Near duplicates: same text differing by a few tokens

Examples of near duplicates: one comma difference

- ①: Terms of service and licenses
- ②: Formulaic writing (copy/pasted or generated from a template)
- ③: Minor formatting differences in copy/pasting